

11.1: Graphs from Parametric Equations:

Parametric equations use a third variable to express the other 2 variables. This makes more complex graphs possible.
The 3rd Variable is the parameter.

To plot a graph, you can either:

⇒ Find Coordinates of many points by adjusting the parameter & sketch.
⇒ Use the graphical calculator setting (curvy)
⇒ Eliminate the parameter, and sketch the Cartesian graph.

11.2: Eliminating the Parameter:

For a given circle, a point has value:

$\Rightarrow \sin\theta = \frac{y}{r}$, $\Rightarrow \cos\theta = \frac{x}{r}$

$\Rightarrow x = r\cos\theta$, $y = r\sin\theta$

This is the basic parametric equation of a circle.
For centre (a, b)

$\Rightarrow x = a + r\cos\theta$, $y = b + r\sin\theta$

From here, a Cartesian Equation is trivially derived

To find a Cartesian Equation, eliminate the parameter.
This can be done using:

⇒ Manipulation to remove the parameter
⇒ Trig identities
⇒ Differentiation & Integrating

11.3: Parametric Differentiation:

According to the chain rule:

$\Rightarrow \frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

Hence finding $\frac{dy}{d\theta} / \frac{dx}{d\theta}$, where θ is a parameter, allows $\frac{dy}{dx}$ to be found.

From here, it is generally advised to use:

$\Rightarrow y - y_1 = \frac{dy}{dx}(x - x_1)$

To find an equation of a line, because other forms are more difficult.